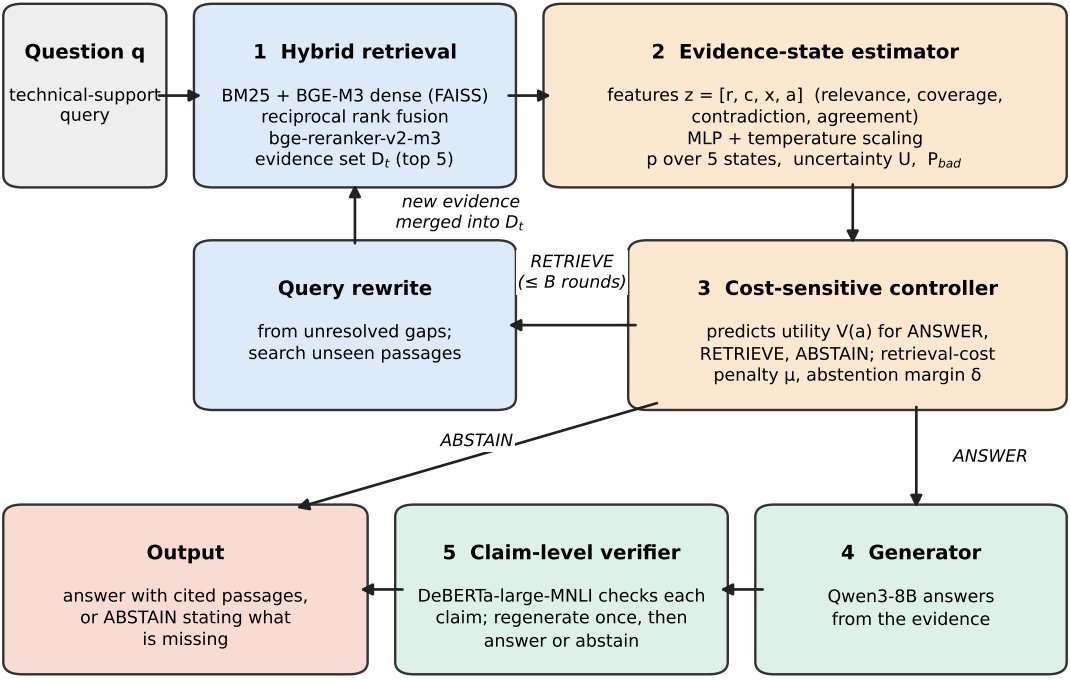
Question q
technical-support query
1 Hybrid retrieval
BM25 + BGE-M3 dense (FAISS)
reciprocal rank fusion
bge-reranker-v2-m3
evidence set D_t (top 5)
2 Evidence-state estimator
features z = [r, c, x, a] (relevance, coverage, contradiction, agreement)
MLP + temperature scaling
p over 5 states, uncertainty U, P_{bad}
new evidence merged into D_t
RETRIEVE (≤ B rounds)
Query rewrite
from unresolved gaps; search unseen passages
3 Cost-sensitive controller
predicts utility V(a) for ANSWER, RETRIEVE, ABSTAIN; retrieval-cost penalty μ, abstention margin δ
ABSTAIN
ANSWER
Output
answer with cited passages, or ABSTAIN stating what is missing
5 Claim-level verifier
DeBERTa-large-MNLI checks each claim; regenerate once, then answer or abstain
4 Generator
Qwen3-8B answers from the evidence